

**BHARGAVI DAMA**  
**SENIOR DATA ENGINEER**

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**PROFESSIONAL SUMMARY:**

- Over **6 years of experience** in building and managing large-scale **data engineering solutions** with a strong emphasis on **cloud-based platforms**, modern **ETL/ELT frameworks**, and **real-time data processing**.
- Extensive experience in **AWS ecosystem**, including **S3, Glue, Redshift, Athena, EMR, Lambda, RDS, Kinesis, DMS, IAM, and CloudWatch** for designing secure, cost-effective, and high-performing data pipelines.
- Designed and developed end-to-end **ETL/ELT pipelines** using **AWS Glue, Step Functions, and Lambda**, handling complex business logic and large-scale data transformations.
- Strong hands-on experience with **Databricks, Delta Lake, and PySpark**, processing batch and streaming datasets across data lakes and lakehouse architectures.
- Built **distributed data processing workflows** using **Apache Spark (PySpark)** and **EMR**, with effective use of partitioning, caching, and job optimization techniques to improve performance
- Designed and deployed **real-time data pipelines** using **Apache Kafka, Kafka Streams, and AWS Kinesis**, enabling near real-time analytics and event-driven data ingestion.
- Developed scalable **ETL processes** using **Python, SQL, and Informatica**, ensuring data quality, validation, and transformation across multiple source systems.
- Experienced in building and optimizing **data warehouses** using **Amazon Redshift, Snowflake, and PostgreSQL**, applying **dimensional modeling (star/snowflake schemas)**, indexing, and clustering for reporting efficiency.
- Automated data workflows using **Terraform, Docker, and CI/CD pipelines**, ensuring smooth deployment across dev, test, and prod environments
- Proficient in writing advanced **SQL and SnowSQL** queries for data extraction, transformation, validation, and performance tuning in complex analytical environments.
- Developed **CI/CD pipelines** using **GitHub Actions, Jenkins, and Docker** to automate testing and deployment of Spark jobs, Glue scripts, and SQL workflows.
- Implemented robust **monitoring and alerting** solutions using **AWS CloudWatch, Databricks job metrics, and CloudTrail**, enabling proactive issue detection and resolution.
- Ensured **data governance, compliance, and security** by applying **role-based access control (RBAC), encryption**, and audit logging in AWS and Snowflake environments.
- Comfortable working in cross-functional teams, translating business requirements into scalable data engineering solutions.
- Experienced in Agile/ Scrum and waterfall methodologies , adapting project approaches to meet diverse requirements .

## TECHNICAL SKILLS:

|                                    |                                                                                    |
|------------------------------------|------------------------------------------------------------------------------------|
| <b>Programming Languages</b>       | Python, R, SQL, Scala                                                              |
| <b>Big Data Technologies</b>       | Hadoop, Spark, Hive, Kafka, AWS EMR, Azure HDInsight                               |
| <b>Clouds</b>                      | AWS, Azure                                                                         |
| <b>Database Management</b>         | MySQL, PostgreSQL, SQL Server, Oracle, MongoDB                                     |
| <b>ETL/ELT Tools</b>               | Informatica Power Centre, AWS Glue                                                 |
| <b>Business Intelligence Tools</b> | Tableau, Power BI, OBIEE, AWS Quick Sight                                          |
| <b>Data Modeling</b>               | Data Warehousing, Mapping, Validation, Integrity, Governance, Stewardship          |
| <b>Data Integration</b>            | ETL/ELT Processes, Data Pipelines, Data Ingestion, Data Transformation             |
| <b>Real-Time Analytics</b>         | Spark Streaming, Kafka Streams, Spark SQL, Kafka                                   |
| <b>Data Quality and Governance</b> | Informatica Data Quality, Metadata-Driven Approaches, CI/CD Pipelines              |
| <b>Scripting Languages</b>         | Shell scripting, R, Python (pandas, NumPy)                                         |
| <b>Monitoring Solutions</b>        | Performance Monitoring (Scala applications, Python scripts), Logging, Audit Trails |

## PROFESSIONAL EXPERIENCE:

**Client:** US Bank, OH.

**Aug 2023 – Present**

**Role:** Sr. Data Engineer

### Responsibilities:

- Designed and developed scalable **ETL/ELT** pipelines using PySpark in Databricks to process high-volume transactional and customer data into structured, analytics-ready formats.
- Tuned **PySpark** jobs and **Databricks** notebooks with memory management, caching strategies, and optimized partitioning, improving batch performance in critical reporting pipelines.
- Engineered **AWS Glue**-based data pipelines integrated with **Lambda** and Step Functions to automate ingestion and transformation of financial data, reducing processing time by 40%
- Implemented real-time data ingestion and transformation using **Snowpipe**, Streams, and Tasks in **Snowflake**, enabling near real-time fraud detection and financial analytics.
- Automated data ingestion pipelines using **AWS Glue Studio** and **PySpark** to reduce operational overhead and improve data reliability for risk and compliance reporting.
- Developed **SQL**-based transformation logic and schema management using **SnowSQL** in Snowflake, optimizing query performance for high-frequency trading and regulatory dashboards.
- Built reusable **Python** modules for **ETL** logic, API integrations (e.g., financial market data), and schema evolution across pipeline stages.
- Engineered real-time streaming pipelines using Spark Structured Streaming and **Kafka** to support anti-money laundering (AML) systems and transaction monitoring use cases.
- Built secure and scalable cloud data architectures using **AWS** services like **S3**, **Glue**, **Lambda**, **Redshift**, and Step Functions for regulatory-compliant data pipelines.
- Developed scalable data models and materialized views in **Redshift** to support dashboarding needs for financial KPIs, reducing BI query latency.
- Integrated Great Expectations into **CI/CD pipelines** for data quality validation, ensuring clean and accurate reporting for compliance and audit teams.
- Automated nightly checks using Python and BOTO3 to verify pipeline success, reducing manual validation efforts by 70% and ensuring timely report generation.
- Developed and maintained complex SQL scripts and stored procedures in Snowflake, Redshift, and Synapse to support financial reconciliation and ledger transformations.

- Designed API-based integrations using **AWS Lambda** and Databricks to ingest third-party financial feeds, with automated reconciliation and validation logic in Python.
- Monitored and optimized **ETL** workloads using **AWS CloudWatch**, **CloudTrail**, and **Redshift Query Monitor** for continuous system reliability and governance.
- Collaborated with data analysts, compliance officers, and DevOps teams to deliver end-to-end financial data solutions using Databricks' shared workspace and version control tools.
- Created Ansible playbooks to manage **AWS** infrastructure configurations for **ETL** nodes, enabling seamless deployment and rollback during critical data operations.
- Utilized environment variables within **Docker containers** for dynamic configuration, making it easier to adapt to different environments and scenarios.

**Environment:** AWS (S3, Glue, Lambda, EMR, Redshift, RDS, IAM, KMS, CloudWatch, CloudTrail, Step Functions), Apache Spark, PySpark, Apache Kafka, Apache Airflow, PostgreSQL, MongoDB, Databricks, Python, SQL, NoSQL, Git, Jenkins, CI/CD, Docker, Amazon QuickSight, Tableau, REST APIs

**Client:** Liberty Mutual Insurance

**Nov 2019 – Dec 2021**

**Role:** Sr. Data Engineer

**Responsibilities:**

- Built and managed **ETL pipelines using AWS Glue**, transforming and loading data from diverse sources into **Redshift and RDS** for analytics and reporting.
- Designed and orchestrated **complex data workflows in Apache Airflow**, ensuring end-to-end automation of data ingestion, transformation, and validation.
- Utilized **AWS EMR** clusters to process large-scale data transformations with **Apache Spark**, optimizing resource usage and performance.
- Developed **Spark-based pipelines in Python** for processing structured and unstructured data, including customer behavior and usage analytics.
- Scaled and tuned **Spark infrastructure** to support high-volume batch workloads with consistent SLAs.
- Implemented **event-driven data processing** using **AWS Lambda**, triggering downstream actions upon S3 uploads or Glue job completion.
- Maintained comprehensive **documentation** for AWS Glue and Lambda functions, ensuring knowledge sharing and audit readiness.
- Built **real-time dashboards in Amazon QuickSight**, sourcing data from **Redshift and RDS** to deliver actionable insights.
- Developed robust **data validation and cleansing routines** within Glue jobs to ensure high data quality and consistency.
- Wrote and optimized **complex SQL queries, stored procedures, and triggers in PostgreSQL** to meet reporting and analytics requirements.
- Integrated **Kafka with logging systems** for centralized log tracking and auditing of streaming data pipelines.
- Implemented **Kafka security** measures including data encryption in transit and access control audits.
- Built **monitoring tools in Python** to track data pipeline health, detect anomalies, and send alerts for proactive issue resolution.
- Managed **Python dependencies and environments** across EMR, Glue, and Lambda using pip and virtualenv for seamless deployments.
- Developed **NoSQL solutions using MongoDB** to handle schema-less, semi-structured data models for flexible storage and retrieval.

**Environment:** AWS (Glue, EMR, Lambda, S3, RDS, Redshift, DynamoDB, QuickSight), Apache Airflow, Apache Spark, Kafka, Python, PostgreSQL, SQL, MongoDB, Linux

**Client: Thomson Reuters, Chennai, India.**

**April 2018–Nov 2019**

**Role: Data Engineer**

**Responsibilities:**

- Developed robust **ETL pipelines** using **Python, PostgreSQL, and AWS Glue** to automate extraction, transformation, and loading of structured and semi-structured data.
- Created and optimized **SQL queries, reports, and data models** in **SQL Server** and **Amazon Redshift**, ensuring fast and reliable analytics.
- Improved **query performance** in Redshift and Athena using data partitioning, compression, and distribution techniques.
- Utilized **AWS Glue Data Catalog** and metadata-driven workflows to automate data integration, reducing manual effort and improving scalability.
- Designed and deployed scalable **Python-based data solutions**, enabling the processing of high-volume datasets across distributed systems.
- Maintained clear and concise documentation for Python scripts to support team collaboration and reduce onboarding time.
- Built **real-time data pipelines** using **Kafka and Scala**, enabling efficient streaming and transformation of operational data.
- Conducted **PySpark job monitoring and optimization**, ensuring performance and system reliability in distributed processing environments.
- Implemented **disaster recovery plans** for AWS Glue, Redshift, and RDS using automated snapshots and backups.
- Enforced **data governance and quality** practices within **NoSQL and Hadoop environments** to ensure consistency and compliance.
- Conducted audits for Hadoop-based workflows to maintain regulatory alignment and minimize risk.
- Implemented **logging and alerting mechanisms** in MongoDB to proactively identify performance issues.
- Developed **Hadoop automation scripts** to streamline repetitive data tasks and improve team efficiency.
- Applied **security measures** including access controls, encryption, and authentication across **AWS, Hadoop, and Kafka**, aligning with internal governance policies.
- Delivered curated datasets and dashboards for strategic business use cases such as reporting, regulatory compliance, and operational analytics.

**Environment:** AWS (Glue, Redshift, S3, RDS, Athena), Python, SQL Server, PostgreSQL, PySpark, Kafka, Scala, Hadoop (HDFS, Hive), MongoDB, NoSQL

**EDUCATION:**

**UNIVERSITY OF NORTH TEXAS , Masters in Computer science**

**QIS COLLEGE OF ENGINEERING AND TECHNOLOGY , Bachelor's in Computer science and Engineering .**